

Joan Alcaide-Nunez

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Last updated: April 2026

EDUCATION

Ludwig-Maximilians-Universität (LMU) Munich

Oct 2025 – Present

B.Sc. in Physics

German School of Barcelona

Sept 2017 – May 2025

Dual High School Diploma (Germany + Spain), GPA: 1.0/1.0

Additional Training:

- *Gravity and Black Holes* — Perimeter Institute (GoPhysics!), 2025
- *Mathematics, Statistics, and Data Science* — Autonomous University of Barcelona (UAB), 2022
- *Algorithms and Programming (C++)* - Polytechnic University of Catalonia (UPC), 2022
- League of Codes (C++) - Harbour Space, 2022-2023

RESEARCH EXPERIENCE

GRB Cosmography & Relativistic Foundations

June – July 2025

Osservatorio Astronomico di Brera (OAB-INAF), Italy

- **Project:** Constrained cosmological density parameters (Ω_m, Ω_Λ) via χ^2 minimization of empirical $E_{\text{peak}} - E_{\text{iso}}$ relation and explored synergies with simulated binary neutron star merger populations from Einstein Telescope expectations. Link to the [repository](#).
- **Mentorship:** Received training by Dr. Gabriele Ghisellini about the physics of GRBs and special relativity effects.
- *Supervisors:* Dr. Giancarlo Ghirlanda, Dr. Om Sharan Salafia

PI GAMMA Project (CAPIBARA Collaboration)

2024 – Present

- Leading a team of students ([GAMMA](#)) at the [CAPIBARA Collaboration](#) currently studying the scientific motivation and requirements for a student-led γ -ray transient observing CubeSat mission.
- I founded the CAPIBARA Collaboration in 2024 with the aim of forming a group of science enthusiastic high school students (some are already in university) for sharing our little knowledge and interests, get to know each other and learn together.

Type Ia Supernovae: Mapping the Expansion History

June – July 2024

Institute of Space Sciences (ICE-CSIC-IEEC), Spain

- **Project:** Computed luminosity distances for Type Ia SNe using infrared open imagery from ESO (SOFI) and photometry data from ZTF and ATLAS using the `sncosmo` library.
- **Formalism:** Learn about cosmological modeling beyond the linear redshift-distance relation, including the Friedmann Equations and fitting density parameters.
- *Supervisor:* Dr. Lluís Galbany

Cepheid Variable Stars Research

Aug – Dec 2023

Youth & Science Program

- **Project:** Compute the Hubble constant ($H_0 = 70.26 \pm 7.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$) replicating Hubble's and Lemaître's methods with modern Cepheid (Konkoly Observatory) and redshift data (NASA-IPAC NED). Link to [report](#).
- **First time:** First exposure to rigorous scientific research and writing within the first year of the Youth and Science fellowship.

- *Supervisors: Dr. Ignasi Pérez-Ràfols, Dr. Laia Casamiquela*

Student Internship at German Space Agency (DLR)

Oct 2024

Institute for Remote Sensing (DLR-IMF), Germany

- Worked with photogrammetry departments on hyperspectral imaging and LandSat thermal analysis.

AWARDS & HONORS

Youth & Science Fellowship: 3-year funded fellowship for research stays internationally.

Abiturpreis German Physical Society (DPG): Physics Distinction for Academic Achievement in high school.

IAAC: Silver Honour and National Award for Spain (2025), top 7% of 12,000 students. Similar awards in 2023 and 2024.

Jugend Forscht: First Prize Iberia (2024/2025), First prizes in regional and national rounds, Special Award in Scientific Photography (NRW 2025).

TALKS & OUTREACH

SSEA Symposium: The CRD Instrument and the COSMOS program, April 2026.

CosmoXarxa: High-energy astrophysics presentation at Barcelona Science Museum, March 2025.

Invited Talk: Intro to Astrophysics for 11th graders, German School Barcelona, 2024.

Outreach: Space Exploration Activity for 3rd grades, Escola Canigó Public School, 2023.

Explainers (Volunteer): Science outreach at CosmoCaixa Museum, 2022–2023.

Slides of most talks can be found on my website.

WORK EXPERIENCE

Ludwig-Maximilians-Universität (LMU) Munich: Student job at central diversity management and research funding unit.

SKILLS

Programming: Python (*numpy*, *scipy*, *pandas*, *matplotlib*, *astropy*), LaTeX, Markdown.

Tools: Git/GitHub, Visual Studio Code, NeoVim, Linux environment.